

The Effects of School Capital Investments: A Reassessment^{*}

Saannidhya Rawat[†]

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Abstract

This paper extends [Biasi, Lafortune and Schönholzer \(2025\)](#), who estimate the effects of school capital investments on student achievement and house prices using a stacked dynamic regression discontinuity design based on local bond elections. First, I independently reproduce their main first-stage and reduced-form estimates. Next, I audit the design’s close-election logic by re-estimating the model under increasingly narrow vote-margin windows: the medium-run effects lose significance as the near-threshold sample collapses and standard errors widen several-fold, yet the restricted-window confidence intervals still contain the published estimates—suggesting a loss of statistical power for the near-threshold observations. Finally, I disaggregate the national estimates by state and find that the dispersion of local returns is large relative to the average; national level estimates mask substantial state-level heterogeneity that I interpret descriptively by relating these effects to the baseline capital stock, student composition and project mix of each state.

Keywords: School capital investment, education finance, regression discontinuity, state heterogeneity, school bonds

JEL Codes: H75, I22, I28, R31

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[†]PhD Candidate, University of Cincinnati. Corresponding author: rawatsa@mail.uc.edu

1 Introduction

Capital investments in schools are a central part of local education finance. Districts use bond elections to renovate buildings, replace aging infrastructure, expand capacity, and improve learning environments. Whether these projects improve student achievement and neighborhood values is an empirical question with direct policy relevance: capital projects are expensive, locally financed, and often justified by both educational and housing-market returns. A long line of research has tried to recover their causal effects but has presented mixed evidence. [Cellini, Ferreira and Rothstein \(2010\)](#) pioneered the use of a dynamic regression discontinuity (RD) design around close bond elections in California, finding that authorized spending raises house prices but has muted measured achievement effects; [Neilson and Zimmerman \(2014\)](#) and [Martorell, Stange and McFarlin Jr \(2016\)](#) reach mixed conclusions on test scores using construction timing and facility-condition variation; and [Hong and Zimmer \(2016\)](#) and [Rauscher \(2020\)](#) document positive but uneven effects across districts and student subgroups. These results sit within a broader school-spending literature in which causal estimates are, on average, positive but heterogeneous ([Jackson, Johnson and Persico, 2016](#); [Lafortune, Rothstein and Schanzenbach, 2018](#); [Baron, 2022](#); [Abott et al., 2020](#)); heterogeneity that the recent meta-analysis of [Jackson and Mackevicius \(2024\)](#) shows is the rule rather than the exception.

[Biasi, Lafortune and Schönholzer \(2025\)](#), hereafter BLS, provide the most comprehensive recent estimates of these effects. Using school bond elections across the United States from 1995 to 2017, they develop a stacked dynamic regression discontinuity design that compares districts around state-specific bond-passage thresholds while accounting for repeated elections and bond histories. Their main results show that bond authorization increases capital spending, raises test scores over the medium run, and is capitalized into house prices, and that these average returns mask substantial variation by project type and district characteristics. Because the paper is both methodologically ambitious and policy-relevant, it is a natural candidate for the kind of post-publication scrutiny that [Hamermesh \(2007\)](#), [Clemens](#)

(2017), and Christensen and Miguel (2018) argue should be routine in economics but remains rare (Chang and Li, 2015; Galiani, Gertler and Romero, 2017; Duvendack, Palmer-Jones and Reed, 2017).

In this paper, I replicate and extend the work of Biasi, Lafortune and Schönholzer (2025). I re-run the original Stata workflow and obtain the published first-stage and reduced-form estimates almost exactly, and I confirm them with an independently coded implementation in R. In the taxonomy of Clemens (2017), this is a successful *verification* that the headline tables can be regenerated from the replication package with high fidelity.

Next, I assess the robustness of the design’s “close-election” logic. The published specification uses the full analysis sample with flexible polynomial controls for the running variable; identification of a local effect at the threshold therefore rests on functional-form extrapolation away from the cutoff, the well-known tension in RD practice between efficiency and locality (Lee and Lemieux, 2010; Calonico, Cattaneo and Titiunik, 2014; Imbens and Kalyanaraman, 2012). I treat the full-sample estimates as the baseline and re-estimate the model under increasingly narrow vote-margin windows. As the window tightens the near-threshold sample collapses, standard errors widen several-fold, and the medium-run effects lose significance; but the restricted-window confidence intervals continue to contain the published estimates. I interpret these results as the signature of loss of power near the threshold rather than evidence against the design, consistent with the broad validity of close-election RD documented by Eggers et al. (2015) and the cautions of Caughey and Sekhon (2011) and de la Cuesta and Imai (2016).

The third contribution assesses external validity by disaggregating the national estimates by state. Biasi, Lafortune and Schönholzer (2025) already report rich heterogeneity by project type and district demographics; my aim is complementary and explicitly generalization-oriented. A single national average is a precision-weighted aggregate of heterogeneous local effects (Goodman-Bacon, 2021; Sun and Abraham, 2021; Callaway and Sant’Anna, 2021), so its policy relevance for any one state depends on how much the underlying effects vary—

precisely the “contexts of convenience” problem that [Handel and Hanushek \(2024\)](#) and [Allcott \(2015\)](#) raise for school-finance evidence. I therefore report state-specific effects and read their dispersion as a map of where the national result is more and less likely to travel, relating it to descriptive state-level summaries of project mix, baseline capital stock, and student composition. Because the number of states is small and state characteristics are not randomly assigned, this mapping is descriptive rather than causal, and I interpret the per-state estimates as a distribution of noisy effects in the spirit of [Meager \(2019\)](#).

The paper is organized as follows. Section 2 characterizes the original study, data, and replication design. Section 3 reports the computational reproduction. Section 4 presents the vote-margin robustness audit. Section 5 reports the state-level heterogeneity and external-validity evidence. Section 6 concludes.

2 Original Study and Replication Design

[Biasi, Lafortune and Schönholzer \(2025\)](#) ask whether and for whom school capital investments financed through local bond elections improve student achievement and are valued by households. Methodologically, they assemble the first near-national panel of bond referenda linked to district finances, test scores, and house prices, and they identify dynamic effects with a stacked regression discontinuity design built around state-specific approval thresholds. Substantively, they find that bond passage raises capital spending, improves test scores over the medium run, and is capitalized into house prices on average, with returns that differ by project type and district characteristics. This finding matters because school capital is a large, locally financed, and recurrent expenditure whose returns had previously been estimated mainly within single states ([Cellini, Ferreira and Rothstein, 2010](#); [Hong and Zimmer, 2016](#); [Rauscher, 2020](#)); a credible national estimate is directly relevant to education-finance policy. This replication reproduces the central first-stage and reduced-form estimates, then tests their robustness to local close-election comparisons and their external validity across

states.

2.1 Data

The analysis relies on the national panel of school capital elections and outcomes constructed by [Biasi, Lafortune and Schönholzer \(2025\)](#). The original authors assemble a database of school bond referenda from administrative records and public information requests, covering local bond elections in 41 U.S. states between 1995 and 2017. After the sample restrictions used in their analysis, the main replication sample contains 29 states.

The bond election data are linked to annual district-level finances from the NCES Annual Survey of School System Finances and the Census of Governments, student achievement data from the Stanford Education Data Archive, and district-level housing price indices from [Contat and Larson \(2024\)](#). The resulting panel allows the original paper to estimate fiscal, academic, and housing-market responses to capital projects authorized through local bond elections.

I do not alter the original data construction for the computational replication. For the state-level extension, I use the same stacked files but estimate the main specifications separately by state after applying the same McCrary-based state screen and the additional state-level sample restrictions documented in Appendix Table 6.

2.2 Stacked Dynamic RD Framework

The original study exploits discontinuities in bond authorization generated by state-specific approval thresholds. Let m_{jc} denote the vote margin in district j and election cohort c , defined as the difference between the yes-vote share and the required approval threshold. A bond is authorized when $m_{jc} \geq 0$. Outcomes Y_{jct} include capital expenditures per pupil, standardized test scores, and a district-level house price index.

Because districts can hold multiple elections over time, [Biasi, Lafortune and Schönholzer \(2025\)](#) use a stacked dynamic regression discontinuity design. Each election is treated as a

cohort, and districts are followed in event time around that election. The stacked panel is indexed by district, cohort, and calendar year. The estimating equation includes district-by-cohort fixed effects, state-by-year fixed effects, event-time treatment indicators, flexible functions of the vote margin, and controls for other bond elections in the district’s history.

In compact form, the main OLS specification is

$$Y_{jct} = \alpha_{jc} + \gamma_{s(j)t} + \sum_{\tau \neq 0} \beta_{\tau} \mathbf{1}\{t - c = \tau\} \mathbf{1}\{m_{jc} \geq 0\} + X'_{jct} \delta + g_c(m_{jc}) + u_{jct},$$

where α_{jc} are district-by-cohort fixed effects, $\gamma_{s(j)t}$ are state-by-year fixed effects, X_{jct} contains bond-history controls, and $g_c(m_{jc})$ denotes cohort-specific vote-margin polynomials. Standard errors are clustered at the district level.

Two features of this estimand are worth making explicit, because they motivate the robustness audit in Section 4. First, the β_{τ} identify a treatment effect at the threshold under the standard RD continuity and no-manipulation conditions (Lee, 2008; Lee and Lemieux, 2010), whose credibility in close-election settings is supported by the smoothness of the running-variable density (McCrary, 2008; Cattaneo, Jansson and Ma, 2020) and by the broad empirical validity of close-election designs documented in Eggers et al. (2015). Second, because $g_c(\cdot)$ is fit on the full sample, the local effect is recovered in part by extrapolating the vote-margin polynomials toward the cutoff rather than by comparison among near-threshold elections alone—the efficiency-versus-locality trade-off emphasized by Calonico, Cattaneo and Titiunik (2014) and Imbens and Kalyanaraman (2012). The full-sample specification buys precision at the cost of leaning on functional form, which is exactly the assumption a margin-restriction audit probes.

3 Computational Replication of Main Results

3.1 First Stage

Table 1 presents the replication of the first-stage effects of bond authorization on school expenditures reported in Table II of [Biasi, Lafortune and Schönholzer \(2025\)](#). The Stata replication reproduces the original estimates exactly, matching point estimates and standard errors across the expenditure categories. The independent R implementation yields nearly identical estimates, with only trivial numerical differences. The first-stage results confirm that bond passage leads to substantial increases in capital expenditures in years 1–5, followed by a decline as projects move toward completion.

Table 1: Replication of BLS Table II: First Stage Effects on Expenditures

Panel A. Capital expenditure			
Estimate	BLS	Stata	R
1-5 years	325*** (93)	325*** (93)	327*** (93)
6-8 years	-239** (104)	-239** (104)	-207* (106)
9-10 years	2 (110)	2 (110)	46 (110)
Panel B. Current expenditure			
Estimate	BLS	Stata	R
1-5 years	7 (30)	7 (30)	-6 (30)
6-8 years	-32 (47)	-32 (47)	-42 (47)
9-10 years	-64 (51)	-64 (51)	-76 (52)
Panel C. Other non-instructional services			
Estimate	BLS	Stata	R
1-5 years	3 (4)	3 (4)	3 (4)
6-8 years	7 (7)	7 (7)	7 (7)
9-10 years	10 (8)	10 (8)	10 (9)

Notes: Table compares point estimates reported in Table II of [Biasi, Lafortune and Schönholzer \(2025\)](#) (in “BLS” column), estimates produced using the original Stata replication code (in “Stata” column) and the replication code in R (in “R” column). Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.2 Mean Effects

Table 2 replicates the reduced-form effects on test scores and house prices from Table III of [Biasi, Lafortune and Schönholzer \(2025\)](#). Across overall test scores, math, ELA, and

the house price index, the replicated estimates closely match the published results. The computational replication therefore supports the central conclusion of the original study: school bond authorization increases school capital investment and is followed by gains in measured achievement and housing values.

Table 2: Replication of BLS Table III: Mean Effects on Test Scores and House Prices

Panel A. Test scores			
Estimate	BLS	Stata	R
1-4 years	0.051*** (0.014)	0.051*** (0.014)	0.051*** (0.014)
5-8 years	0.096*** (0.023)	0.096*** (0.023)	0.096*** (0.023)
9-12 years	0.091*** (0.029)	0.091*** (0.029)	0.091*** (0.030)
Panel B. Math			
Estimate	BLS	Stata	R
1-4 years	0.054*** (0.018)	0.054*** (0.018)	0.054*** (0.018)
5-8 years	0.096*** (0.028)	0.096*** (0.028)	0.096*** (0.028)
9-12 years	0.068** (0.033)	0.068** (0.033)	0.068** (0.033)
Panel C. ELA			
Estimate	BLS	Stata	R
1-4 years	0.050*** (0.014)	0.050*** (0.014)	0.050*** (0.014)
5-8 years	0.102*** (0.022)	0.102*** (0.022)	0.102*** (0.022)
9-12 years	0.115*** (0.030)	0.115*** (0.030)	0.115*** (0.030)
Panel D. House Price Index			
Estimate	BLS	Stata	R
1-4 years	2.397* (1.287)	2.397* (1.287)	2.397* (1.291)
5-8 years	6.783*** (2.047)	6.783*** (2.047)	6.783*** (2.055)
9-12 years	4.822** (1.927)	4.822** (1.927)	4.822** (1.934)

Notes: Table compares point estimates reported in Table III of [Biasi, Lafortune and Schönholzer \(2025\)](#) (in “BLS” column), estimates produced using the original Stata replication code (in “Stata” column) and the replication code in R (in “R” column). Stars denote significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4 Robustness to Election-Margin Restrictions

A regression discontinuity design identifies a treatment effect *at* the cutoff, but the published specification estimates that effect from the full sample using flexible polynomial controls for the vote margin. Identification of the local effect therefore relies in part on extrapolation of those polynomials toward the threshold from elections decided by wide margins—the stan-

dard efficiency-versus-locality trade-off in RD estimation (Lee and Lemieux, 2010; Calonico, Cattaneo and Titiunik, 2014; Imbens and Kalyanaraman, 2012). A natural robustness check is to ask whether the headline patterns survive when the comparison is forced to be more local. This section reports that check; following Clemens (2017), it is a robustness exercise around the baseline, not a competing estimator.

Figure 1 displays the distribution of vote margins in the replication sample, where the margin is defined as the difference between the share of yes votes and the state-mandated approval threshold. The density is continuous through the threshold, with no pronounced bunching just above zero—the qualitative pattern that formal density tests (McCrary, 2008; Cattaneo, Jansson and Ma, 2020) are designed to detect, and the kind of setting in which close-election RD has been shown to be broadly valid across tens of thousands of races (Eggers et al., 2015). The baseline BLS specification uses this full sample with flexible controls for the running variable and the stacked DRD structure described above. I preserve that specification as the benchmark and estimate additional versions that restrict the sample to elections within 20, 15, 10, and 5 percentage points of the threshold.

Narrower windows move the comparison closer to the RD cutoff but sharply reduce the sample. Appendix Table 8 shows that moving from the full sample to a ± 5 -percentage-point window cuts the test-score panel from about 1.05 million to 49 thousand observations and the contributing elections from roughly 10,500 to 4,100, while the treated share falls from 0.85 to 0.58 as the retained elections become more balanced around the threshold. Figure 2 plots the resulting coefficient path across windows for each outcome, and Appendix Table 7 reports the underlying estimates. In principle a change in the estimates as the window tightens admits three readings: the full-sample estimator could be leaning on functional-form extrapolation; the local estimand at the cutoff could genuinely differ from the full-sample average; or precision could simply collapse as the effective sample shrinks. The evidence is most consistent with the third. The medium- and long-run test-score effects, which are positive and significant in the full sample (0.038 and 0.047 standard deviations), attenuate toward

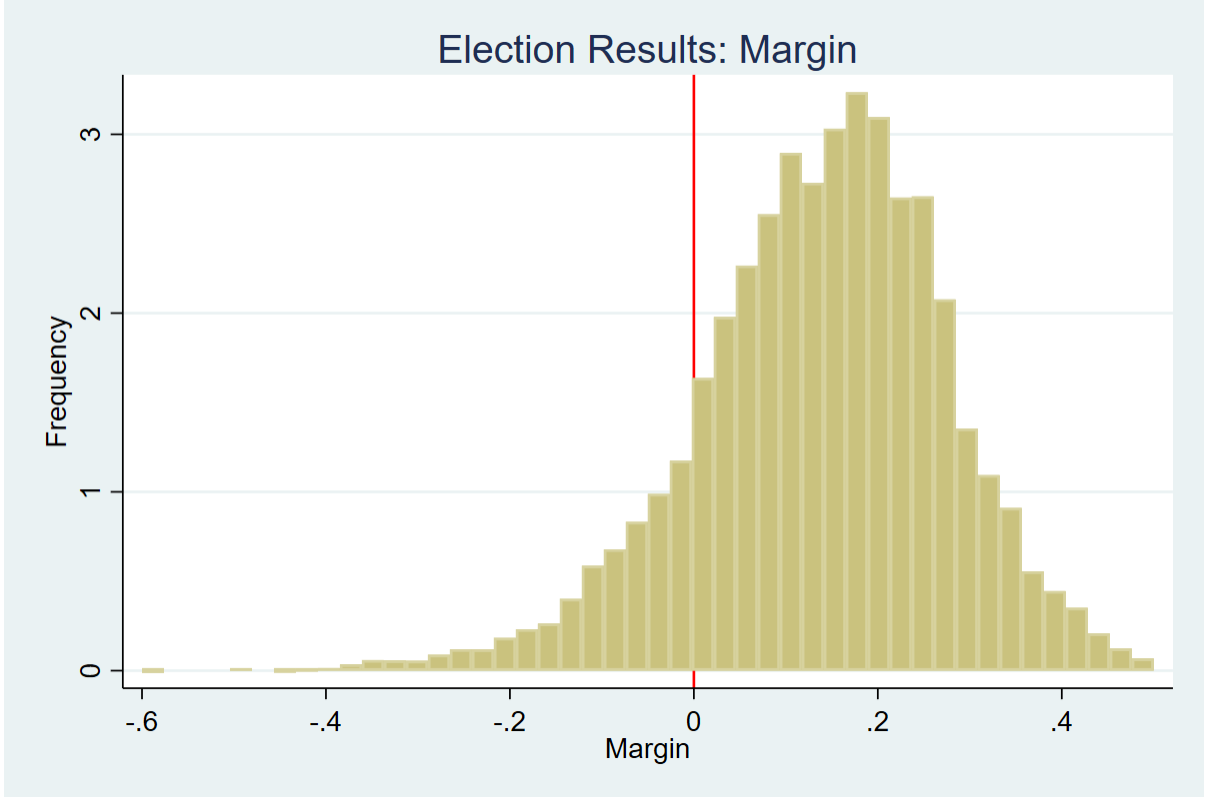


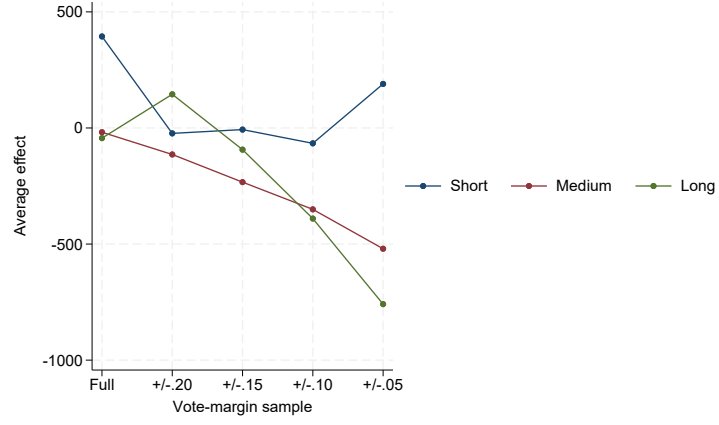
Figure 1: School Bond Election Vote Margins

zero and lose significance as the window narrows, with standard errors widening three- to five-fold (from about 0.011 to 0.055); at the two tightest windows the point estimates even turn slightly negative. The first-stage capital-spending effect behaves the same way, strongly positive in the full sample but insignificant once the sample is restricted. Crucially, these restricted-window estimates are too imprecise to separate the positive baseline from zero: their confidence intervals generally contain the full-sample estimate, so the near-threshold data neither overturn nor independently confirm the dynamic effects. If anything, the house-price effects *sharpen* at intermediate windows, turning positive and significant at the ± 15 - and ± 20 -point bandwidths. Two features argue against reading this instability as a design failure rather than a power loss: the running-variable density is smooth through the threshold (Figure 1), the pattern that formal manipulation tests are built to detect (McCrary, 2008; Cattaneo, Jansson and Ma, 2020), and close-election RD designs are broadly valid in settings like this one (Eggers et al., 2015; de la Cuesta and Imai, 2016). The audit thus

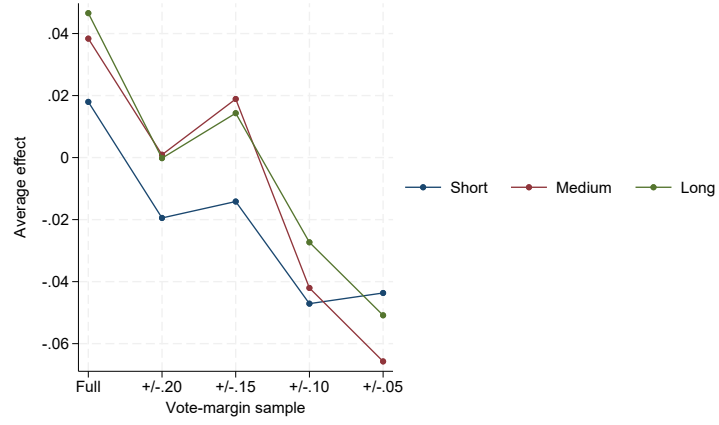
leaves the original conclusions standing while making their basis transparent: by using information away from the cutoff, the full-sample specification is far better powered to detect the dynamic effects than any near-threshold subsample (Imbens and Kalyanaraman, 2012; Calonico, Cattaneo and Titiunik, 2014). A fully nonparametric local-polynomial implementation with data-driven bandwidths and formal density diagnostics (Calonico et al., 2017; Cattaneo, Jansson and Ma, 2020) is the natural way to push the local analysis further and is left to future work.

5 State-Level Heterogeneity and Descriptive Mechanisms

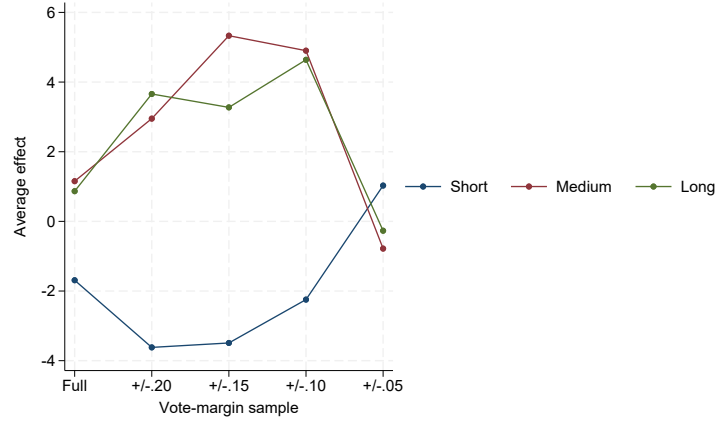
The national estimates in Biasi, Lafortune and Schönholzer (2025) summarize the *average* effect of school capital investments, but a national average is only as portable as the effects underlying it are similar. A growing literature shows that design-based estimates of school-spending effects vary widely across settings (Jackson and Mackevicius, 2024; Handel and Hanushek, 2024), and that an average estimated from the contexts that happen to enter a study need not describe the policy’s return elsewhere (Allcott, 2015). Because a stacked dynamic design recovers a precision-weighted aggregate of underlying cohort- and state-specific effects (Goodman-Bacon, 2021; Sun and Abraham, 2021; Callaway and Sant’Anna, 2021), decomposing that aggregate is the natural way to gauge how far the national result generalizes. I therefore re-estimate the stacked dynamic RD specification separately for each state with sufficient usable data, using the same event-time windows as the main replication: years 1–5, 6–8, and 9–10 for capital spending, and years 1–4, 5–8, and 9–12 for test scores and house prices. Because per-state estimates are individually noisy, I read the collection of state effects as a *distribution* of returns rather than as a ranking of states, in the spirit of the hierarchical treatment of heterogeneous estimates in Meager (2019, 2022). Figure 3 collects the state-specific estimates for all three outcomes.



(A) Capital spending per pupil



(B) Test scores



(C) House price index

Figure 2: Coefficient Path Under Vote-Margin Restrictions

Notes: Each panel traces the average short-, medium-, and long-run effect across the full sample (baseline) and samples restricted to within 20, 15, 10, and 5 points of the threshold; see Appendix Table 7 for the underlying estimates and exact horizon definitions.

5.1 Capital Spending

Panel (A) of Figure 3 shows that bond authorization does not translate into identical capital-spending responses across states. Several states show large short-run increases in capital expenditures per pupil, while others show smaller or imprecise responses. This heterogeneity matters for interpretation: later test-score and housing-price effects are mediated by whether authorized bonds actually generate new capital investment and by the timing of that spending.

5.2 Test Scores

Panel (B) of Figure 3 reports state-specific reduced-form effects on standardized test scores, and it is the central piece of evidence on external validity. The confirmed national effect is positive, but the state-level estimates are spread on both sides of it: a number of states show positive medium- or long-run effects, while others are near zero or negative, and the dispersion across states is large relative to the national point estimate. Two states can therefore both be consistent with the national average while implying very different local returns to a capital project. This is the same lesson Cummins (2017) draws in revisiting an influential education experiment—an informative average can coexist with subgroups for which the effect is small or reversed—and it is the quantitative content of the “contexts of convenience” caution of Handel and Hanushek (2024): the national estimate describes the policy’s return in the average reporting state, not in any particular one. Consistent with the meta-analytic evidence on school spending (Jackson and Mackevicius, 2024), the message is not that the original average is wrong, but that it should be used as a distribution of plausible state-level returns rather than as a single transportable number.

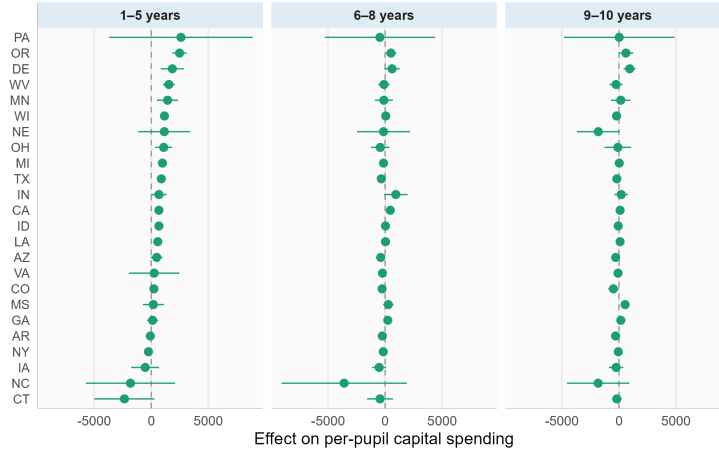
5.3 Housing Market Responses

Panel (C) of Figure 3 shows a similarly heterogeneous pattern for housing prices. In some states, house prices rise after bond authorization, consistent with capitalization of expected improvements in school quality or local amenities. In other states, the estimates are small, negative, or imprecise. These patterns are consistent with the possibility that housing-market responses depend on both the perceived benefits of capital projects and the fiscal costs associated with bond repayment.

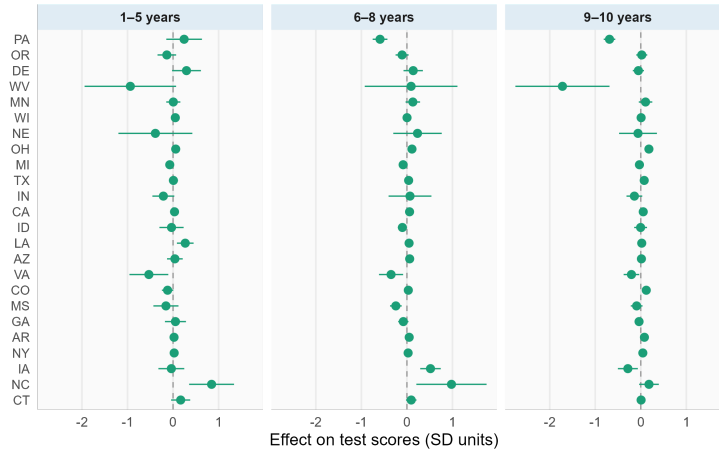
5.4 Descriptive Mechanism Evidence

A natural next question is whether the cross-state dispersion lines up with the mechanisms emphasized by [Biasi, Lafortune and Schönholzer \(2025\)](#): the type of capital project, the quality of the pre-existing capital stock, and the characteristics of students served. I link the state-level estimates to descriptive state averages from the replication package—project-category shares, baseline capital stock, free or reduced-price lunch share, and minority share—while stressing that these variables are correlates, not causal instruments, for state heterogeneity, since states are few and were not randomly assigned their project mix or demographics ([Allcott, 2015](#)).

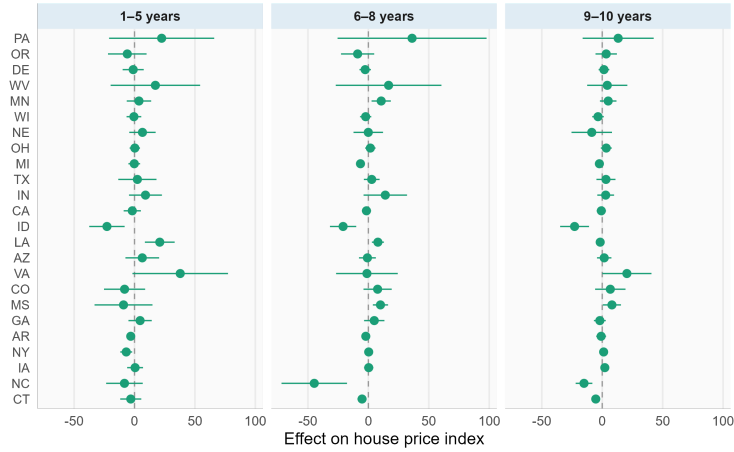
Appendix Figure 4 plots the medium-run test-score effects against these mechanism variables, and the appendix tables report the underlying summaries and bivariate associations. The associations are weak and imprecise: with roughly two dozen noisy state estimates and incomplete project-category coverage, the data cannot adjudicate among the candidate mechanisms. I therefore treat them as hypothesis-generating rather than as confirmatory evidence on the mechanisms. Their value is to identify which contexts—project types, capital-stock conditions, and student populations—most warrant the properly powered, design-based mechanism analysis that would be needed to explain why the return to capital investment generalizes well to some states and not others, the kind of work that [Handel and Hanushek \(2024\)](#) argue the school-finance literature most needs.



(A) Capital expenditures per pupil



(B) Test scores



(C) House price index

Figure 3: Effect of School Bond Authorization by State

Notes: Markers are state-specific stacked dynamic RD point estimates and whiskers are district-clustered confidence intervals; underlying values and exact event-time horizons appear in Appendix Tables 3-5.

6 Conclusion

This paper starts by independently regenerating the original first-stage and reduced-form estimates of [Biasi, Lafortune and Schönholzer \(2025\)](#). Next, it extends the analysis by restricting the sample to different vote-margin windows around the decision threshold. I find that the medium-run effects lose significance as the near-threshold sample collapses, yet the restricted-window confidence intervals still contain the published estimates—a power loss near the threshold rather than a contradiction of the design. Finally, the state-level decomposition shows that the national average is a weighted blend of heterogeneous local returns in capital spending, test scores, and house prices.

Taken together, these results strengthen the original study while developing a more nuanced understanding of the effects. The dispersion across states is a quantitative statement of the “contexts of convenience” cautionary tale of [Handel and Hanushek \(2024\)](#): a national average estimated from the states that happen to hold and report close bond elections need not describe the return to capital in a given state. The descriptive associations between state effects and project mix, baseline capital stock, and student composition are weak and are not causal evidence about institutions or project categories. Instead, they highlight where additional data collection and a properly powered, design-based mechanism analysis would be valuable for understanding why the return to capital investment generalizes well to some states and not others.

Data and Code Availability

Author-written code will be made available as supplementary material and archived in a public repository. Access to the underlying data follows the terms of the Biasi, Lafortune and Schönholzer replication package and its third-party data providers.

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Appendix

A.1 State-Level Estimate Tables

Table 3: Effect of School Bond Authorization on Capital Spending by State

State	Capital spending (per-pupil dollars)		
	1–5 years	6–8 years	9–10 years
AZ	478.2 (267.3)	-388.3*** (94.8)	-291.9*** (82.6)
AR	-77.9 (125.9)	-242.2** (114.0)	-293.4 (186.9)
CA	670.9*** (134.3)	455.9*** (79.8)	99.8 (95.8)
CO	231.4 (190.0)	-264.8 (175.9)	-486.8** (232.0)
CT	-2350.4* (1341.6)	-442.0 (581.8)	-187.7 (219.5)
DE	1853.0*** (521.7)	607.2* (347.0)	937.7*** (269.5)
GA	116.7 (250.9)	235.3** (114.1)	160.4 (143.4)
ID	669.5*** (143.2)	36.2 (75.7)	-67.9 (114.5)
IN	677.1** (340.1)	934.5* (531.9)	198.2 (303.5)
IA	-529.0 (624.7)	-519.8* (308.6)	-250.6 (321.7)
LA	573.0*** (210.8)	34.5 (69.1)	96.3 (97.5)
MI	981.9*** (172.5)	-140.0 (93.6)	24.9 (101.5)
MN	1432.8*** (477.9)	-111.2 (406.5)	153.9 (437.1)
MS	183.7 (472.1)	276.1 (233.4)	539.4*** (188.5)
NE	1145.6 (1162.1)	-138.6 (1187.3)	-1811.8* (961.3)
NY	-245.5 (194.4)	-162.9 (128.2)	-63.0 (139.9)
NC	-1820.0 (1998.0)	-3590.2 (2801.1)	-1828.3 (1391.6)
OH	1094.1*** (379.9)	-426.4 (410.9)	-97.5 (596.4)
OR	2482.2*** (325.4)	515.4** (237.4)	607.5* (328.9)
PA	2605.8 (3208.0)	-458.7 (2463.1)	16.7 (2475.1)
TX	883.5*** (163.4)	-338.2** (149.3)	-188.7 (136.4)
VA	259.3 (1124.9)	-231.9 (186.1)	-85.9 (192.9)
WV	1542.5*** (272.4)	-101.8 (256.2)	-257.5 (293.1)
WI	1156.3*** (185.1)	53.0 (97.9)	-211.5** (103.1)

Notes: Entries report state-specific treatment effects of passing a school bond on per-pupil capital spending over the indicated post-election horizons. Standard errors in parentheses. Horizon bins follow the specification in Biasi, Lafor-tune, and Schönholzer (2025). *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 4: Effect of School Bond Authorization on Test Scores by State

State	Test scores (SD units)		
	1–4 years	5–8 years	9–12 years
AZ	0.040 (0.088)	0.061 (0.047)	0.014 (0.038)
AR	0.022 (0.027)	0.050* (0.027)	0.080*** (0.029)
CA	0.032 (0.033)	0.058*** (0.015)	0.054*** (0.019)
CO	-0.120* (0.064)	0.031 (0.036)	0.120*** (0.031)
CT	0.167 (0.107)	0.094 (0.059)	0.007 (0.038)
DE	0.295* (0.162)	0.138 (0.108)	-0.053 (0.063)
GA	0.053 (0.118)	-0.079 (0.058)	-0.038 (0.030)
ID	-0.034 (0.136)	-0.100** (0.051)	-0.004 (0.072)
IN	-0.213* (0.123)	0.068 (0.240)	-0.141 (0.088)
IA	-0.038 (0.145)	0.518*** (0.115)	-0.284** (0.112)
LA	0.268*** (0.094)	0.047 (0.037)	0.021 (0.031)
MI	-0.072** (0.032)	-0.083*** (0.028)	-0.030* (0.018)
MN	0.006 (0.079)	0.130 (0.081)	0.104 (0.076)
MS	-0.156 (0.141)	-0.243*** (0.066)	-0.091 (0.066)
NE	-0.388 (0.414)	0.233 (0.272)	-0.061 (0.212)
NY	0.025 (0.024)	0.026 (0.018)	0.045* (0.023)
NC	0.846*** (0.252)	0.978** (0.393)	0.180 (0.110)
OH	0.058 (0.044)	0.111*** (0.031)	0.178*** (0.049)
OR	-0.136 (0.104)	-0.106 (0.073)	0.021 (0.061)
PA	0.243 (0.200)	-0.590*** (0.083)	-0.687*** (0.065)
TX	0.008 (0.051)	0.038 (0.034)	0.075** (0.036)
VA	-0.531** (0.218)	-0.348*** (0.135)	-0.205** (0.087)
WV	-0.938* (0.513)	0.092 (0.519)	-1.721*** (0.527)
WI	0.050 (0.039)	0.003 (0.035)	0.007 (0.021)

Notes: Entries report state-specific treatment effects of passing a school bond on test scores (in within-state standard deviation units) over the indicated post-election horizons. Standard errors in parentheses. Horizon bins follow the specification in Biasi, Lafortune, and Schönholzer (2025). *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 5: Effect of School Bond Authorization on House Prices by State

State	House price index		
	1–4 years	5–8 years	9–12 years
AZ	6.43 (7.12)	-0.80 (3.57)	1.63 (3.04)
AR	-3.12* (1.80)	-2.13 (1.63)	-0.91 (2.07)
CA	-1.85 (3.64)	-1.59 (1.93)	-0.77 (1.96)
CO	-8.22 (8.69)	7.64 (5.96)	6.68 (6.42)
CT	-3.10 (4.45)	-5.23*** (1.93)	-5.37*** (1.22)
DE	-1.10 (4.45)	-2.65 (2.41)	1.43 (2.25)
GA	4.63 (4.92)	4.86 (4.30)	-2.00 (2.50)
ID	-22.81*** (7.47)	-20.92*** (5.55)	-22.90*** (6.10)
IN	9.08 (6.94)	14.00 (9.18)	2.85 (3.54)
IA	0.48 (3.35)	0.28 (1.97)	2.14* (1.20)
LA	20.90*** (6.29)	7.88*** (2.53)	-1.70 (1.71)
MI	-0.19 (2.48)	-6.54*** (1.71)	-2.37 (1.59)
MN	3.68 (5.16)	10.61*** (4.07)	4.92 (3.50)
MS	-9.10 (12.23)	10.00*** (3.21)	8.01** (3.82)
NE	6.58 (5.57)	-0.08 (6.23)	-8.69 (8.55)
NY	-6.84*** (2.52)	0.27 (0.97)	1.15 (1.00)
NC	-8.32 (7.72)	-44.78*** (13.83)	-15.09*** (3.53)
OH	0.27 (2.19)	1.65 (2.20)	3.33 (2.31)
OR	-5.98 (8.15)	-8.90 (7.01)	3.28 (4.51)
PA	22.47 (22.19)	36.16 (31.46)	13.18 (15.00)
TX	2.37 (8.07)	2.76 (3.32)	3.07 (4.02)
VA	37.84* (20.18)	-1.20 (13.02)	20.38** (10.38)
WV	17.27 (18.90)	16.73 (22.31)	4.15 (8.48)
WI	-0.49 (3.11)	-2.25 (2.38)	-3.50 (2.44)

Notes: Entries report state-specific treatment effects of passing a school bond on local house price index over the indicated post-election horizons. Standard errors in parentheses. Horizon bins follow the specification in Biasi, Lafortune, and Schönholzer (2025). *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

A.2 State Sample Audit

Table 6: State Sample Audit for the State-Level Extension

FIPS	State	Included	Districts	Elections	Observations	Status
5	AR	1	184	1463	19007	Included
4	AZ	1	122	242	3091	Included
6	CA	1	675	1110	13937	Included
8	CO	1	130	205	2612	Included
9	CT	1	22	37	489	Included
10	DE	1	16	37	452	Included
13	GA	1	158	499	6642	Included
19	IA	1	161	225	2782	Included
16	ID	1	84	154	2035	Included
18	IN	1	52	58	633	Included
22	LA	1	41	100	1250	Included
26	MI	1	406	684	8846	Included
27	MN	1	202	288	2964	Included
28	MS	1	55	63	818	Included
37	NC	1	13	15	131	Included
31	NE	1	67	85	1072	Included
36	NY	1	306	2422	30047	Included
39	OH	1	423	715	7773	Included
41	OR	1	141	220	2657	Included
42	PA	1	13	14	190	Included
48	TX	1	669	1174	13685	Included
51	VA	1	18	39	549	Included
55	WI	1	326	539	6753	Included
54	WV	1	32	47	667	Included
12	FL	0	1	1	16	Limited sample
25	MA	0	231	2984	36980	McCrary screen
24	MD	0	2	13	158	Limited sample
29	MO	0	327	761	9955	McCrary screen
38	ND	0	15	23	225	Limited sample
32	NV	0	14	19	234	Limited sample
40	OK	0	166	386	4946	McCrary screen
44	RI	0	5	5	46	Limited sample
53	WA	0	246	646	7640	McCrary screen

Notes: The included states follow the 24-state sample reported in the state-level extension. Missouri, Washington, Oklahoma, and Massachusetts are excluded by the McCrary screen used in the BLS analysis. Florida, Maryland, Nevada, North Dakota, and Rhode Island are excluded from the reported state extension because the usable state-level sample is too limited for the separate stacked DRD estimates or is not retained in the reported state-effect output.

A.3 Vote-Margin Robustness Checks

Table 7: Robustness to Vote-Margin Restrictions

Bandwidth	Outcome	Horizon	Estimate	SE	N
full	capital	1-5	394.260***	(74.500)	141341
full	capital	6-8	-18.235	(46.038)	141341
full	capital	9-10	-43.865	(44.011)	141341
full	hpi	1-4	-1.690	(1.667)	94625
full	hpi	5-8	1.154	(0.956)	94625
full	hpi	9-12	0.867	(1.017)	94625
full	test	1-4	0.018	(0.016)	1047701
full	test	5-8	0.038***	(0.013)	1047701
full	test	9-12	0.047***	(0.011)	1047701
bw20	capital	1-5	-23.229	(136.083)	24917
bw20	capital	6-8	-114.101	(137.251)	24917
bw20	capital	9-10	145.105	(116.137)	24917
bw20	hpi	1-4	-3.619**	(1.743)	17872
bw20	hpi	5-8	2.951*	(1.534)	17872
bw20	hpi	9-12	3.658**	(1.601)	17872
bw20	test	1-4	-0.019	(0.019)	236481
bw20	test	5-8	0.001	(0.020)	236481
bw20	test	9-12	-0.000	(0.023)	236481
bw15	capital	1-5	-6.890	(158.105)	16805
bw15	capital	6-8	-233.274	(177.622)	16805
bw15	capital	9-10	-93.323	(177.756)	16805
bw15	hpi	1-4	-3.491*	(1.951)	12011
bw15	hpi	5-8	5.332***	(1.884)	12011
bw15	hpi	9-12	3.273*	(1.915)	12011
bw15	test	1-4	-0.014	(0.019)	170763
bw15	test	5-8	0.019	(0.026)	170763
bw15	test	9-12	0.014	(0.028)	170763
bw10	capital	1-5	-66.075	(226.770)	9578
bw10	capital	6-8	-350.558	(241.661)	9578
bw10	capital	9-10	-390.152	(242.927)	9578
bw10	hpi	1-4	-2.246	(2.328)	6676
bw10	hpi	5-8	4.903*	(2.858)	6676
bw10	hpi	9-12	4.639	(2.819)	6676
bw10	test	1-4	-0.047*	(0.026)	110136
bw10	test	5-8	-0.042	(0.031)	110136
bw10	test	9-12	-0.027	(0.033)	110136
bw05	capital	1-5	189.787	(406.930)	3122
bw05	capital	6-8	-520.193	(349.106)	3122
bw05	capital	9-10	-758.741	(471.676)	3122
bw05	hpi	1-4	1.029	(3.012)	2055
bw05	hpi	5-8	-0.782	(3.603)	2055
bw05	hpi	9-12	-0.269	(5.216)	2055
bw05	test	1-4	-0.044	(0.044)	49439
bw05	test	5-8	-0.066	(0.053)	49439
bw05	test	9-12	-0.051	(0.056)	49439

Notes: The full-sample row is the BLS baseline stacked DRD design. Other rows restrict the sample to elections with absolute vote margins below the listed threshold. Standard errors are clustered at the district level. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

Table 8: Sample Retention Under Vote-Margin Restrictions

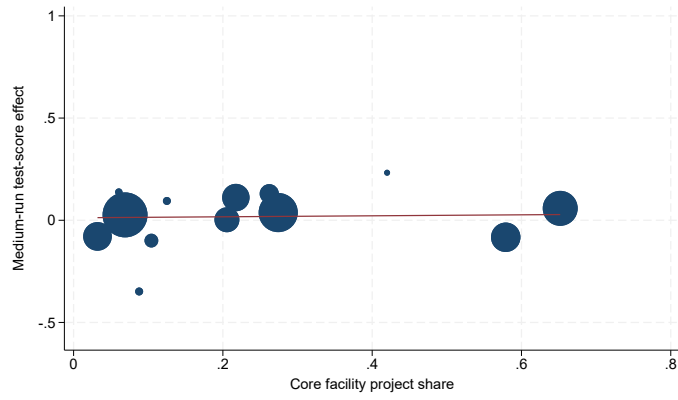
Bandwidth	Observations	Elections	Districts	Treated obs.	Pass rate	Margin range
full	1133519	10496	4353	360370	0.854	[-0.500, 0.500]
bw20	259942	9216	3785	206632	0.795	[-0.200, 0.200]
bw15	188607	8122	3349	140820	0.747	[-0.150, 0.150]
bw10	123161	6511	2721	83736	0.680	[-0.100, 0.100]
bw05	58146	4105	1729	33525	0.577	[-0.050, 0.050]

A.4 Descriptive Mechanism Evidence

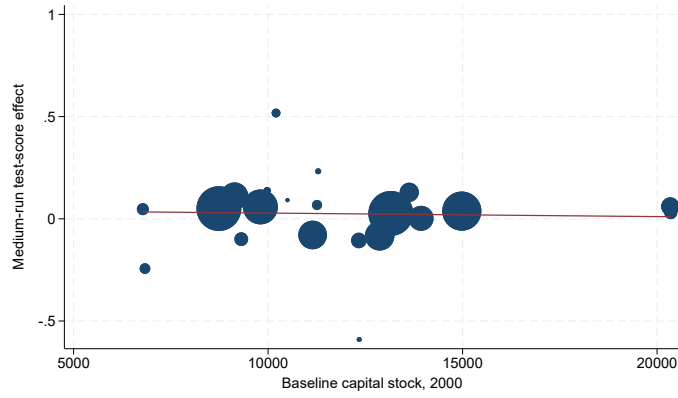
Table 9: State-Level Effects and Descriptive Mechanism Variables

FIPS	Test 5–8	HPI 5–8	Capital 1–5	Core facility share	FRPL share	Capital stock
4	0.061	-0.802	478	.	.	20339
5	0.050	-2.132	-78	.	0.400	8735
6	0.058	-1.587	671	0.652	0.459	9802
8	0.031	7.639	231	.	0.248	20352
9	0.094	-5.230	-2350	0.125	0.232	.
10	0.138	-2.650	1853	0.061	0.309	9976
13	-0.079	4.863	117	0.032	0.394	11144
16	-0.100	-20.924	669	0.104	0.276	9310
18	0.068	14.002	677	.	0.248	11258
19	0.518	0.285	-529	.	0.228	10205
22	0.047	7.877	573	.	0.548	6780
26	-0.083	-6.537	982	0.579	0.275	12869
27	0.130	10.614	1433	0.262	0.220	13629
28	-0.243	10.000	184	.	0.590	6835
31	0.233	-0.083	1146	0.420	0.253	11288
36	0.026	0.269	-245	0.069	0.341	13155
37	0.978	-44.781	-1820	.	0.342	.
39	0.111	1.652	1094	0.217	0.249	9137
41	-0.106	-8.905	2482	.	0.291	12333
42	-0.590	36.162	2606	.	0.289	12343
48	0.038	2.759	883	0.274	0.416	14980
51	-0.348	-1.197	259	0.088	0.265	.
54	0.092	16.727	1543	.	0.436	10498
55	0.003	-2.249	1156	0.205	0.202	13933

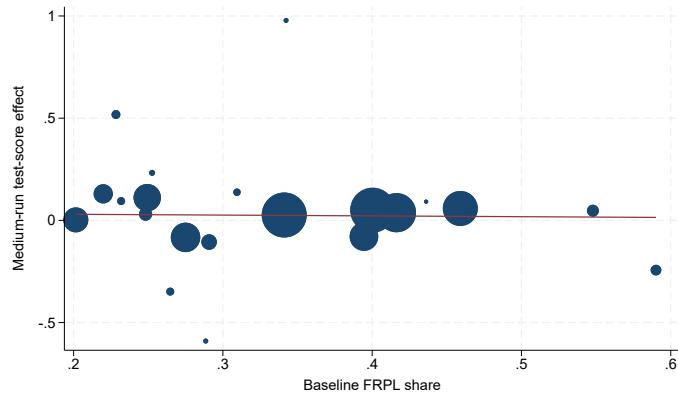
Notes: Mechanism variables are state-level descriptive averages from the BLS replication files. Core facility share averages classroom, STEM, HVAC, infrastructure, and safety/health project indicators. These associations are descriptive and are not interpreted as causal estimates.



(A) Core facility project share



(B) Baseline capital stock



(C) Baseline FRPL share

Figure 4: State-Level Medium-Run Test-Score Effects and Candidate Mechanisms

Notes: Each panel plots state-specific medium-run test-score effects (vertical axis) against a state-level mechanism summary (horizontal axis), with a fitted bivariate regression line. The associations are descriptive, not causal.

Table 10: Descriptive Associations with State-Level Effects

Outcome	Mechanism	Slope	SE	p-value	N
bmed test	core facility share	0.024	0.096	0.804	13
bmed test	noninstructional share	-0.150	0.121	0.243	13
bmed test	frpl share	-0.039	0.234	0.869	23
bmed test	minority share	-0.043	0.143	0.766	23
bmed test	capstock 2000	-0.000	0.000	0.781	21
blong test	core facility share	0.002	0.084	0.978	13
blong test	noninstructional share	-0.119	0.108	0.291	13
blong test	frpl share	0.020	0.229	0.931	23
blong test	minority share	0.035	0.142	0.810	23
blong test	capstock 2000	-0.000	0.000	0.745	21
bmed hpi	core facility share	-7.065	6.190	0.278	13
bmed hpi	noninstructional share	-13.224	7.903	0.122	13
bmed hpi	frpl share	8.308	13.823	0.554	23
bmed hpi	minority share	5.882	7.451	0.439	23
bmed hpi	capstock 2000	0.000	0.000	0.437	21
blong hpi	core facility share	-1.523	6.022	0.805	13
blong hpi	noninstructional share	-2.311	8.136	0.782	13
blong hpi	frpl share	2.761	10.762	0.800	23
blong hpi	minority share	2.082	4.766	0.667	23
blong hpi	capstock 2000	0.000	0.000	0.193	21
bshort cap	core facility share	1360.203	713.049	0.083	13
bshort cap	noninstructional share	-256.354	1109.557	0.822	13
bshort cap	frpl share	-2.4e+03	1652.979	0.165	23
bshort cap	minority share	-1.3e+03	1039.867	0.241	23
bshort cap	capstock 2000	0.040	0.048	0.415	21